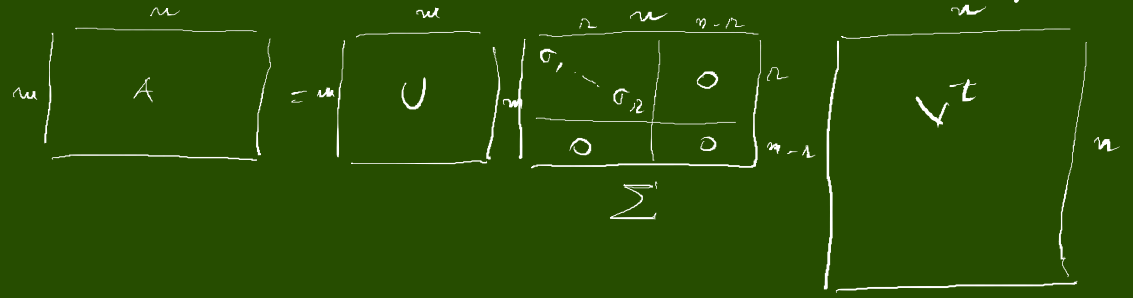
$A \in \mathcal{M}_{m,n}(\mathbb{R}) \Rightarrow \exists\, U, \Sigma, V$ matrice a.i.

$$A = U \Sigma V^t$$
$$(m,n) \quad (m,m)(m,n)$$

$U \in \mathcal{M}_m(\mathbb{R})$ ortogonală

$\Sigma \in \mathcal{M}_{m,n}(\mathbb{R})$

$V \in \mathcal{M}_n(\mathbb{R})$ ortogonală



Algoritm

1) Determinarea matricei V.

$$A^tA \qquad (A^tA)^t = A^t(A^t)^t = A^tA$$
$$(n,m)(m,n)$$

$\Rightarrow A^tA$ simetrică $\Rightarrow \sigma(A^tA) \subseteq \mathbb{R}$

$$\Rightarrow \quad A^tA = V \cdot D \cdot V^t$$

ortodiagonalizare

V ortogonală

D diagonală

$V = (v_1 | \dots | v_n)$ $v_1, \dots v_n$ vectori proprii pt A^tA

$D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$ $\lambda_1 \geq \dots \geq \lambda_n$ valori proprii matricei A^tA și

$A^tA v_i = \lambda_i v_i \quad \forall i = \overline{1,n}$

$V = (v_1 | \dots | v_n)$

2) Determinarea matricei Σ

$\lambda_1 \geq \dots \geq \lambda_n \geq 0$

$\lambda_1 \geq \dots \geq \lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_n$

$A^tA v_i = \lambda_i v_i \quad / \cdot v_i^t$

$v_i^t A^t A v_i = \lambda_i v_i^t v_i$

$\langle Av_i, Av_i \rangle \qquad \langle v_i, v_i \rangle$

$\|Av_i\|^2 = \lambda_i \|v_i\|^2 \Rightarrow \lambda_i \geq 0$

$\dim_{\mathbb{R}} \ker(A^tA) = n - r$

$\ker(A^tA) = \ker(A)$ (cursul precedent)

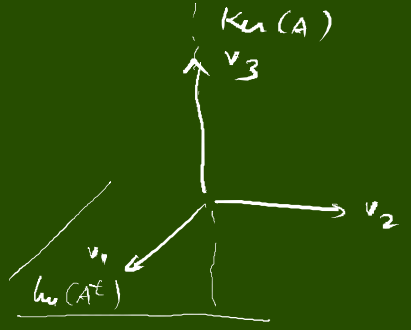
$\Rightarrow \dim \ker(A) = n - r \Rightarrow \dim \operatorname{Im}(A) = r$

$\Rightarrow \operatorname{rang} A = r$.

$\underbrace{v_{r+1}, \dots, v_n}_{\text{ortonormată}} \in \operatorname{Ker}(A^tA) = \operatorname{Ker}(A)$, $\dim \operatorname{Ker} A = n - r$ $\Rightarrow$

$\Rightarrow v_{r+1} \dots v_n$ bază ortonormată în $\operatorname{Ker}(A)$.

$\Rightarrow v_1 \dots v_r$ bază ortonormată în $\operatorname{Ker}(A)^\perp = \operatorname{Im}(A^t)$



$\lambda_1 \geq \dots \geq \lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_n$

$\sigma_i = \sqrt{\lambda_i}$, $i = \overline{1, r}$

valorile singulare ale matricei A

$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$

$$\Sigma = \begin{array}{c} \\ r \\ m-r \end{array} \begin{array}{c} \begin{array}{cc} r & n-r \end{array} \\ \left(\begin{array}{c|c} \begin{matrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_r \end{matrix} & 0 \\ \hline 0 & 0 \end{array}\right) \end{array}$$

3) Determinarea matricei U.

$\|Av_i\|^2 = \lambda_i \|v_i\|^2 \Rightarrow i = \overline{1,n}$

$\|Av_i\| = \sigma_i \cdot \|v_i\| = \sigma_i$

$i = \overline{n+1, m}$ $\quad$ $\|v_i\| = 1$

$Av_i = 0$

$(Av_i)_{i=\overline{1,n}}$ vectori ortogonali

$\langle Av_i, Av_j \rangle = (Av_i)^t Av_j = v_i^t A^t A v_j = \lambda_j \langle v_i, v_j \rangle =$

$A^t A v_j = \lambda_j v_j$

$= \begin{cases} \lambda_i, & i = j \\ 0, & i \neq j \end{cases}$

$i = \overline{1,n}$ $u_i = \frac{1}{\sigma_i} Av_i$ vectori $\Rightarrow$ $u_1, \dots u_n$ vectori ortonormati

Completăm $u_1 \dots u_n \in \mathbb{R}^m$ la o bază ortonormată în $\mathbb{R}^m$

bază ortonormată în $\ker(A^t)$

$u_1, \dots, u_n, u_{n+1}, \dots, u_m$

$\in \operatorname{Im}(A)$

$\dim \operatorname{Im}(A) = n \Rightarrow \{u_1 \dots u_n\}$ bază ortonormată în $\operatorname{Im}(A)$

$\Rightarrow U = (u_1 | \dots | u_m) \in \mathcal{U}_m(\mathbb{R})$ ortogonală

4) $A = U \Sigma V^t \iff AV = U\Sigma$

$$AV = A(v_1 | \dots | v_n) = (Av_1 | \dots | Av_n | 0 | \dots | 0) =$$

$$= (\sigma_1 u_1 | \dots | \sigma_n u_n | 0 | \dots | 0) =$$

$$= \underbrace{(u_1 | \dots | u_n | u_{n+1} | \dots | u_m)}_{U} \cdot \left(\begin{array}{c|c} \sigma_1 & \\ & \ddots \\ & \sigma_n \\ \hline 0 & 0 \end{array}\right)$$

Σ

$$A = U \cdot \Sigma V^t = \underbrace{\sigma_1 u_1 v_1^t}_{(m,1)(1,n)} + \dots + \sigma_n u_n v_n^t$$

matrice de rang 1

$\underline{mn} \leq (n)(m+n)$

$\underline{\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n}$

$$A = U \cdot \Sigma \cdot V^t \iff A^t = V \cdot \Sigma^t \cdot U^t$$

$$\underset{(m,n)(n,m)}{A \cdot A^t} = U \cdot \Sigma \cdot \underbrace{V^t \cdot V}_{I_n} \cdot \Sigma^t \cdot U^t$$

$$= U \cdot \begin{pmatrix} \sigma_1^2 & & & & \\ & \ddots & & & \\ & & \sigma_r^2 & & \\ & & & 0 & \\ & & & & \ddots \\ & & & & 0 \end{pmatrix} U^t \quad \begin{matrix} r \\ m-r \end{matrix} \quad \begin{matrix} r & m-r \end{matrix}$$

$A^t A$	$A A^t$
(n,n)	(m,m)
$n \leq m$	$m \leq n$
$\downarrow$ SVD	$\downarrow$ SVD
A	A^t

Exemplo: $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 2 & 2 \end{pmatrix}$ $(3,2)$

1) $A^t A$ $(2,3)(3,2)$ $(2,2)$

$A A^t$ $(3,2)(2,3)$ $(3,3)$

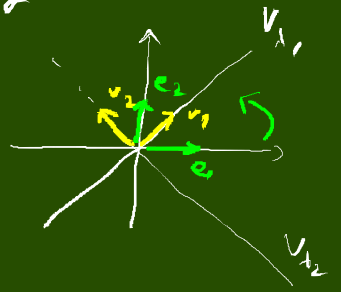
$$A^t A = \begin{pmatrix} 1 & 0 & 2 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}$$

$\Rightarrow \quad \lambda_1 = 10 \geq \lambda_2 = 0$ ~~$A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$~~

$\lambda_1 = 10$: $\begin{cases} 5x + 5y = 10x \\ 5x + 5y = 10y \end{cases} \Rightarrow x = y$

$$V_{\lambda_1} = \left\{ \begin{pmatrix} x \\ x \end{pmatrix} / x \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

$\left\{ v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ base ortonormada por V_{λ_1}



$\lambda_2 = 0$: $V_{\lambda_2} = \left\{ \begin{pmatrix} -y \\ y \end{pmatrix} / y \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$

$\left\{ v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$ base ortonormada por V_{λ_2}

$$\Rightarrow \quad V = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \\ \underbrace{\quad}_{v_1} & \underbrace{\quad}_{v_2} \end{pmatrix}$$

$\Sigma \in \mathcal{M}_{3,2}(\mathbb{R})$ $\quad \Sigma = \left(\begin{array}{c|c} \sigma_1 & 0 \\ \hline 0 & 0 \\ 0 & 0 \end{array}\right) = \left(\begin{array}{c|c} \sqrt{10} & 0 \\ \hline 0 & 0 \\ 0 & 0 \end{array}\right)$

$\lambda_1 = 10 \quad \lambda_2 = 0$ $\quad A = \sigma_1 \cdot u_1 v_1^t$

$\hookrightarrow \sigma_1 = \sqrt{10}$ val. singulare a matricei A

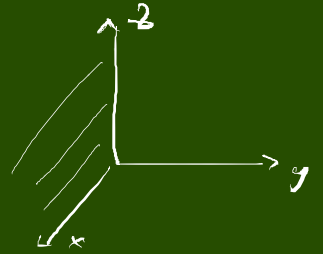
$U = ?$

$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 2 & 2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2\sqrt{5}} \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} =$

$= \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$

Completez u_1 la o baza ortonormata pt $\mathbb{R}^3$

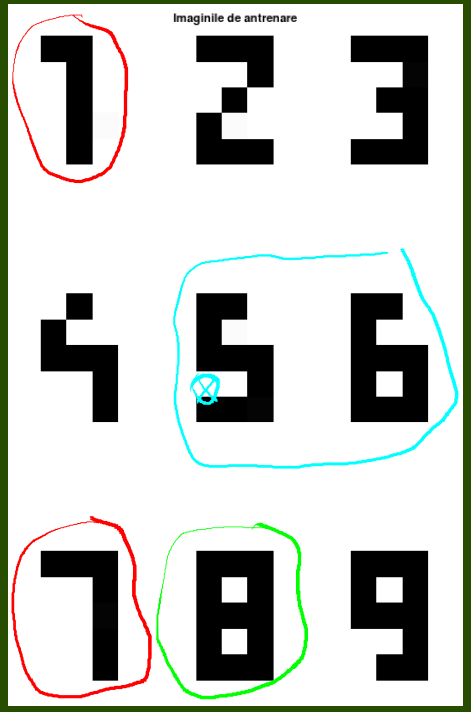
$u_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$, $u_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

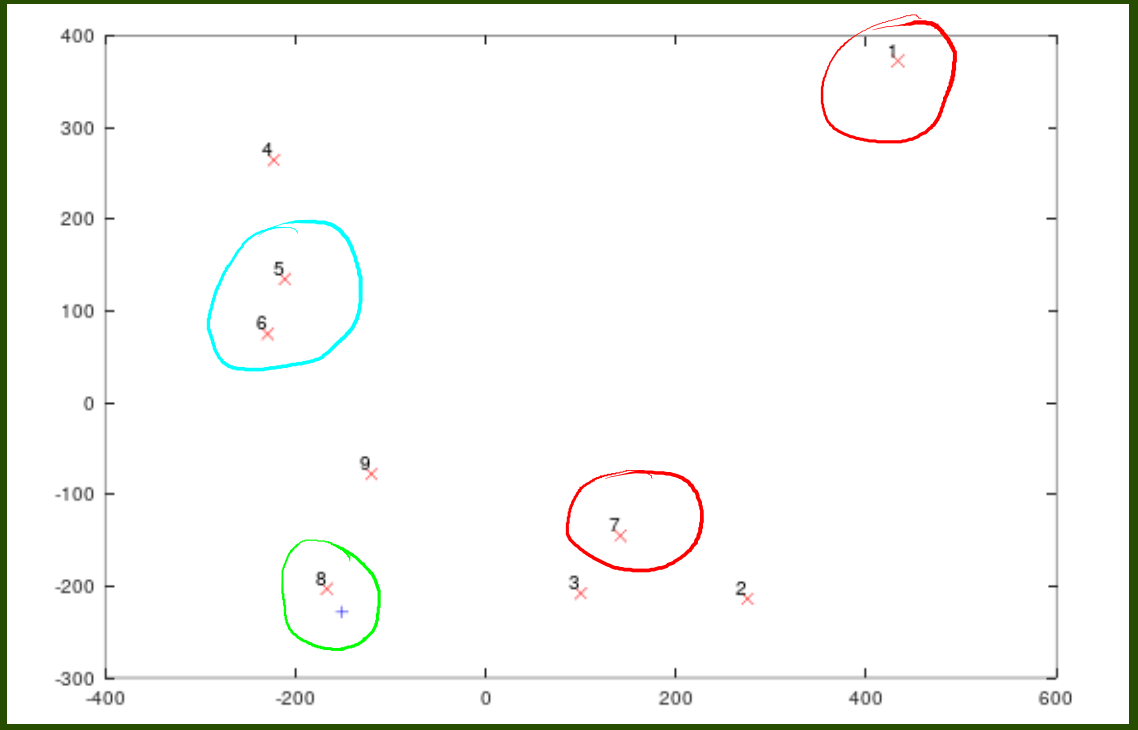


$u_1, u_2 \in \times O_2$

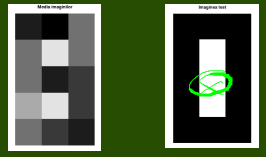
$\Rightarrow U = \begin{pmatrix} 1/\sqrt{5} & 2/\sqrt{5} & 0 \\ 0 & 0 & 1 \\ 2/\sqrt{5} & -1/\sqrt{5} & 0 \end{pmatrix} \Rightarrow A = \underline{U \cdot \Sigma \cdot V^t}$

$u_1 \quad u_2 \quad u_3$





$5 \times 3 \longrightarrow (15,1)$ $x_1 \dots x_9$



μ

$x - \mu$ $\quad p_{U_2}^{\perp}(x-\mu)$ $\quad U_2 = Sp\{u_1, u_2\}$ eigenfaces.

$A = (x_1 - \mu \mid \dots \mid x_9 - \mu) \in M_{15,9}(\mathbb{R}) \longrightarrow A = U \Sigma V^t$ $\quad \sigma_1 \geq \sigma_2 \geq \sigma_3$ $\quad r = \operatorname{rang} A = 3$ $\quad Sp\{u_1, u_2, u_3\}$

(x, y, z) $\quad x^2 + y^2 = z^2 \quad$ $x = u^2 - v^2,\ y = 2uv,\ z = u^2 + v^2$

$u, v \in \mathbb{N}$.

Fermat $\quad x^2 + y^2 = p$ prim $\iff p \equiv 1 \pmod 4$

$A = A^t \qquad q(v) = v^t A v$

$\max_{\|v\|=1} q(v) = \lambda_{max} \qquad \lambda_{max} = \max_{\lambda \in Sp(A)} \lambda$

$\min_{\|v\|=1} q(v) = \lambda_{min}$

$A = Q \cdot D \cdot Q^{-1} \qquad Q \in M_n(\mathbb{R})$ orthog

$D \in M_n(\mathbb{R})$ diag. $D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$

$\lambda_1 \geq \dots \geq \lambda_n$.

$q(v) = v^t A v = \underbrace{(v^t Q)}_{w^t} \cdot D \cdot \underbrace{(Q^t v)}_{w} = (y_1 \dots y_n) \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} =$

$$v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \qquad w = Q^t v = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = [v]_{\mathcal{B}}$$

$$= \underline{\lambda_1 y_1^2 + \dots + \lambda_n y_n^2} \leq \lambda_1 y_1^2 + \dots + \lambda_1 y_n^2 = \lambda_1 \underbrace{(y_1^2 + \dots + y_n^2)}_{1}$$

$$\mathcal{B}_c = \{e_1, \dots e_n\} \xrightarrow{P_{\mathcal{B}_c}^{\mathcal{B}} = Q} \mathcal{B} = \{q_1, \dots q_n\}$$

$$Q = (q_1 | \dots | q_n)$$

$$v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \quad [v]_{\mathcal{B}_c} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \qquad [v]_{\mathcal{B}} = ?$$

$$[v]_{\mathcal{B}_c} = \underbrace{P_{\mathcal{B}_c}^{\mathcal{B}}}_{Q} [v]_{\mathcal{B}} \quad / \cdot Q^t$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = Q^t \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = [v]_{\mathcal{B}}$$

$$\Rightarrow y_1^2 + \dots + y_n^2 = \|v\|^2 = \|Q^t v\|^2 \overset{Q^t \text{ ortog}}{=} \|v\|^2 = 1$$

$$\Rightarrow \quad \xi(v) \leq \lambda_1 \qquad \forall\, v$$

$q(v) = \lambda_1 \iff y_1^2 = 1,\ y_2 = \dots = y_n = 0$

$1 \quad -1$

$\Rightarrow y_1 = 1,\ y_2 = \dots = y_n = 0$

$\Rightarrow [v]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \Rightarrow v = 1 \cdot g_1 + 0 \cdot g_2 + \dots + 0 \cdot g_n$

$= g_1$ vettore proprio rel. λ_1.

$ax^2 + 2bxy + cy^2 + dx + ey + f = 0 \qquad a^2 + b^2 + c^2 \neq 0$

conica

$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{matrix} \lambda_1 \\ \lambda_2 \end{matrix} \qquad \left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_{\mathcal{B}} = \begin{pmatrix} x' \\ y' \end{pmatrix}$

$\lambda_1 x'^2 + \lambda_2 y'^2 + \dots = 0$

$\lambda_1, \lambda_2 > 0 \Rightarrow$ ellisse

$\lambda_1, \lambda_2 < 0 \Rightarrow$ ellisse

$\lambda_1, \lambda_2 = 0 \Rightarrow$ parabola